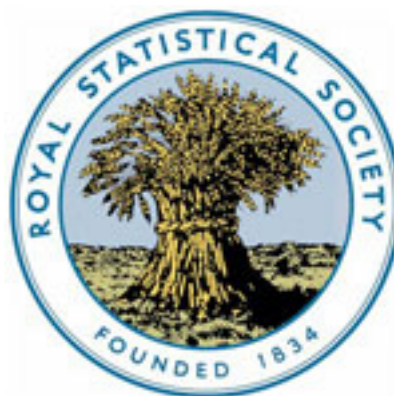




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Reviewed work(s):

Source: *Journal of the Royal Statistical Society. Series D (The Statistician)*, Vol. 31, No. 1, Proceedings of the 1981 I.O.S. Annual Conference on Statistics in Medicine (Mar., 1982), pp. 81-96

Published by: [Blackwell Publishing](#) for the [Royal Statistical Society](#)

Stable URL: <http://www.jstor.org/stable/2988103>

Accessed: 08/07/2012 10:36

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Estimating Hospital Catchment Populations

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Introduction

As a result of the 1974 reorganization of the National Health Service, health authorities were constituted which were responsible both for managing groups of hospitals and for providing community health services for statutory districts. Greater emphasis was to be placed on the integrated managing of health care resources for the benefit of a particular population. Quantitative methods of resource allocation based upon the notion of a target population were devised in an attempt to distribute resources fairly and rationally to the various health authorities in the National Health Service. A problem which this approach encounters, is that the population which looks to a hospital or group of hospitals for in-patient and out-patient services does not always occupy easily defined boundaries and it has been generally recognized that the official resident or statutory populations of health authorities are not the same as the catchment populations of their hospitals.

In this paper a number of different methods of estimating catchment populations are reviewed. On the basis of an analysis of these methods recommendations regarding the preparation of catchment populations estimates are made.

Definition

The following definition of catchment populations is proposed. "For a given hospital or unit and for a particular specialty the catchment population is that group of persons who would attend the hospital or unit were they to require treatment under that specialty". For certain specialties the same person might attend a number of different units depending on

the severity of his condition; in that case either separate catchment populations must be regarded as existing for these various subspecialties or it must be accepted that no simple interpretation of the term “catchment population” is possible.

Methods of Estimation

A number of different methods of estimating catchment populations have been devised and these will be described below.

1. *Bailey's Method (1956, 1977)*

One of the most commonly used methods of estimating catchment populations is that usually known as Bailey's method. It was described by him in an article in 1956 but was also proposed by Norris in 1952. The method is as follows. If we imagine an area Y with population of size n which sends m_1 cases in a given period to hospital X and m_2 cases elsewhere, then the contribution that Y makes to the catchment population of hospital X is estimated to be

$$a = \frac{nm_1}{(m_1 + m_2)}$$

If C is the catchment population then

$$C = \sum a$$

where the summation is over all appropriate areas.

Another way of looking at this calculation is to regard a as being cases produced locally divided by the *local* overall admission rate d . Thus,

$$a = m_1/d$$

where

$$d = (m_1 + m_2)/n$$

This method has been used extensively and there are a number of published studies (Nuffield Provincial Hospitals Trust, 1955, 1962; Barr, 1957; Forsyth and Logan, 1960; Newell, 1964).

2. *The RAWP Method (1976)*

Although the 1976 Resource Allocation Working Party (RAWP) devised a formula which treated the allocation process as a whole (DHSS, 1976), a catchment population element in the formula can be identified (Senn and Shaw, 1978).

The method starts by using the resident or statutory populations. The figure is then adjusted using net cross-boundary flows which are weighted by specialty costs and divided by a cost per head of population. If the cost factors are ignored the method, essentially, is to take net cross-boundary flows, divide these by the *national* admission rate and add the result to the resident population.

If we let

- r = the resident population of the health authority,
- z_1 = the cases from the resident population dealt with elsewhere,
- z_2 = cases from outside the health authority boundaries dealt with by the health authority,
- d^* = the national admission rate,

then the catchment population c is estimated by

$$c = r + \frac{z_2 - z_1}{d^*}$$

3. Gandy's Method (1979)

A similar method to the RAWP approach has been proposed by Gandy (1979). If any cases from the resident populations are treated outside of the home area they are converted to a population equivalent by using an *expected* admission rate. This expected rate is to take into account age and sex characteristics of the resident area as well as making allowance for morbidity by using Standardized Mortality Ratios (SMRs). If any cases are dealt with by the home area hospitals, that have arisen outside of the resident boundaries they are also similarly converted to population equivalents using expected admission rates.

If c , r , z_1 , are as previously defined

- e_1 is the expected admission rate for the resident population,
- e_{2_i} is the expected admission rate in the i th health district from which cases are imported,
- z_{2_i} is the number of cases imported from the i th health district so that $\sum z_{2_i} = z_2$ as defined before,

then

$$c = r + \sum \left(\frac{z_{2_i}}{e_{2_i}} \right) - \frac{z_1}{e_1}$$

where the summation is over all contributing health districts.

4. Cottrell's Method (1979)

This provides yet another variation of the idea of adjusting a resident population using flows of cases.

If we have c , r , z_1 z_2 as previously defined and z_3 =cases from the resident population dealt with by the resident health authority (i.e. un-exported cases) the catchment population is estimated by

$$c = r \left(\frac{z_2 + z_3}{z_1 + z_3} \right)$$

It should be noted that $z_2 + z_3$ is the total of all cases dealt with by the authority and $z_1 + z_3$ is the total of all cases originating in the health authority.

The formula can be equivalently expressed by

$$c = \frac{z_3}{d'} + \frac{z_2}{d'}$$

where

$$d' = \frac{z_1 + z_3}{r}$$

and is the admission rate for the resident population as a whole.

5. Median Line Method (Griffiths, 1972)

Another method which has been used within the health service (e.g. to establish formal catchment boundaries for large psychiatric institutions) is to examine cases admitted from an area to see which hospital is most used. The whole population of that area can then be assigned to that hospital. If every area sends cases to only two hospitals at most, then eventually a boundary may be established forming a "median line" at which 50 per cent of all cases go to a particular authority. Of course there is always the possibility of isolated "islands" of low or high admission proportions occurring. Griffiths (1972) describes a protocol for the calculation of what he terms "equal likelihood service areas".

An example of estimation using these methods is given in an appendix.

Discussion

There are a number of important differences between the five methods which will be discussed below.

First of all, the RAWP, Gandy and Cottrell methods all require the use of a resident population figure and thus to a certain extent beg the question,

since the whole purpose of the exercise is to estimate a population (Senn, 1980, 1981). This “resident population” may be a quite inappropriate starting point for the exercise and it is not always obvious which figure should be used. For example, for a specialty such as plastic surgery housed in a district hospital should we use the district, area or regional population as our starting figure?

Secondly, the Bailey and median line approaches require rather more information than the other methods. Cottrell (1979) has described four kinds of admission data that may be required for catchment calculations, namely: cases arising from the “resident” population and dealt with by the home district, “imports”, “exports”, and finally cases that arise and are dealt with outside the district boundaries but in areas that do contribute some cases to the district hospital. Information on all four categories is required for the Bailey and median line approaches but only information on the first three is required by the other three methods. This is not, however, a very serious objection to Bailey’s method since, if catchment populations are being calculated for a number of health districts, say by a regional management services unit, the fourth category of data will have to be obtained in any case.

Thirdly, all the methods are influenced to a certain extent by the level of service a hospital provides for its catchment population and this is clearly an undesirable feature. This is a point which will be discussed more fully below in terms of a simple model.

Let us assume we take a single geographical unit Y whose population is of size n and can be split into two groups. The first group of size p_1 looks to hospital X for treatment, the second of size p_2 looks elsewhere. So that $n = p_1 + p_2$. The total catchment population of X would be found by summing all such values p_1 over all appropriate geographical units. Let us also assume that the first population produces m_1 cases in a given period and the second m_2 .

Now if we can assume that $m_1/p_1 = m_2/p_2$ (i.e. that admission rates are equal) then we can find

$$p_1 = \frac{nm_1}{m_1 + m_2}$$

and

$$c = \sum \frac{nm_1}{m_1 + m_2}$$

(where the summation is over all contributing geographical areas) which is Bailey’s formula. Bailey’s method can thus be seen to rely on the assump-

tion that admission rates *within* the geographical area are constant. However, admission rates *between* areas do not have to be constant.

Now let us see what assumptions the RAWP method requires. First we assume that the geographical unit *Y* is *within* the health district boundaries; its contribution to the catchment population would be calculated as

$$a = (p_1 + p_2) - \frac{m_2}{d^*}$$

where *a* is an estimate of *p*₁ and *d*^{*} is the national admission rate. If on the other hand unit *Y* is *outside* the health district boundary then

$$a = \frac{m_1}{d^*}$$

Now only if

$$d^* = \frac{m_1}{p_1} = \frac{m_2}{p_2}$$

will the catchment population be correctly estimated. In fact the RAWP approach relies on the assumption of constant admission rates across the country (Senn and Shaw, 1978). Gandy's formula is little better in that it requires that local admission rates are everywhere equal to their expected admission rates using national data. It can be shown similarly that Cottrell's method relies on the following two assumptions:

- (i) that the admission rate across the resident population is constant,
- (ii) that the admission rate in all contributing areas is the same as in the resident population.

The median line method is not easily discussed in the same terms as the other four methods but would appear to be vastly inferior. Except by extreme good fortune a figure estimated on this basis would be unlikely to correspond to the "true" catchment population and no assumption about admission rates alone would guarantee that it did so. Furthermore since an area (say) providing 51 per cent admissions to a hospital *X* receives the same treatment as one providing 100 per cent admissions it would seem to ignore much relevant information.

It is thus clear that providing the data are available, Bailey's method is to be preferred to the other alternatives since it is the method that is least affected by variations in admission rates and requires the least assumptions.

The problem, therefore, of how Bailey's method should be applied will be considered in more detail.

Estimation Using Bailey's Method

The assumption of constant admission rates implicit in Bailey's formula will be more approximately true if small rather than large units of population are used as the basis for the method. It is therefore of interest to know how the variance of the estimate is affected by using small rather than large units for calculation.

Strictly speaking the variance of estimation using Bailey's method is not defined since there will always be a finite probability of obtaining no admissions from a given unit leading to indeterminacy. However in practice we can normally deal with such cases by using admissions from a neighbouring area to apportion population. Bailey gives a formula for the variance of a catchment population based on the assumption that admissions follow a Poisson distribution, which is reasonable providing that the conditions leading to admission are not infectious and there is no waiting list. The formula is

$$\sigma_c^2 \simeq \sum \frac{n^2 m_1 m_2}{m_1 + m_2}^1$$

The formula was derived by Bailey using a Taylor's expansion in two variables to obtain an expression for Δc (Bailey, personal communication).

The same formula may be derived even if admissions are not Poisson using a different set of assumptions. We assume that total admissions from a given area are fixed, with a probability p that a case will be admitted to hospital X .

The proportion of cases observed to go to X will have variance

$$p(1-p)/t$$

where t is the total number of cases. However p can be estimated using $m_1/(m_1 + m_2)$ and $t = m_1 + m_2$. Thus our estimate of the variance of the total catchment population will be as before.

In order to examine the effect of scale on variance one may imagine a catchment calculation based on a single large area over which the admission rate is constant. The population is of size n and m^*_1 cases go to X and m^*_2 cases elsewhere.

We thus have

$$c = \frac{n^* m^*_1}{m^*_1 + m^*_2}$$

$$\sigma_c^2 = \frac{n^{*2} m^*_1 m^*_2}{(m^*_1 + m^*_2)^3}$$

¹ Following Bailey no distinction will be made here between actual and expected admissions which are the Poisson variates and parameters respectively.

If the admission rate is constant across the area and the area is divided into a number of smaller units then the contribution to catchment population of a typical unit is

$$a = \frac{sm_1^*s/n^*}{(m_1^* + m_2^*)s/n^*}$$

where s is the population of this unit and the contribution to variance is, upon substitution $s\sigma_c^2/n^*$.

Since $\sum s = n^*$ the variance of the overall catchment population therefore is the same whether a single large unit or a number of smaller units is used in calculation.

Since Bailey's formula is an approximate one, however, this analysis has been verified using a computer simulation, details of which are given in an appendix.

Application

A problem constantly facing health care planners is the lack of information about the need for health services. This has lead in practice to a number of ingenious attempts to calculate the component elements of need of which the RAWP's efforts are perhaps the most noteworthy. In many cases such ingenuity can be self-defeating. A feature of many of these health care planning calculations is that they are based upon a restricted set of information, the elements of which interact in sometimes unforeseen ways severely affecting the value of the calculations. Here catchment populations are no exception and their method of calculation limits the areas in which they can be usefully applied.

For example if, as a result of a resource study, it is decided to increase provision at a particular site, this may have effects on the estimated catchment population. First, the catchment population itself may actually be effected because increased facilities attract patients who might otherwise have gone elsewhere. Secondly, the estimates themselves will probably be affected. For example, any increase in provision to patients outside the resident area would, using the RAWP or Gandy approach, lead to a corresponding increase in estimated catchment population.

There is, therefore, a dynamic relationship between catchment populations and resource provision (Carstairs and Heasman, 1974). For this reason, catchment populations should be one of many types of management information used in resource allocation and they should be employed cautiously (Skidmore, 1981).

Another use to which catchment populations might be put is to compare admission rates for various health districts. Once again results would have to be interpreted cautiously. The catchment population provides the admissions, but the admissions provide the data used in estimating catchment populations. Furthermore, the estimation methods all make some assumption about areas over which admission rates are constant. There is thus the possibility that resource variation may be underestimated. This sort of difficulty can be overcome to a certain extent, but not completely, by using Bailey's method based on small areas.

If catchment populations are to be used in planning, therefore, they should be used at a tactical level or as a first stage in strategic planning. In deciding where to site a new hospital, for example, existing catchment populations can clearly only have a limited use. They provide useful background information but other factors will have to be considered. Studies may be required on travelling times, hospital waiting lists and also of employment patterns in the local community. In the end political decisions will be required for formulae alone will be inadequate.

Conclusion

A number of simple recommendations are offered in conclusion

- 1 Bailey's method should be used whenever possible.
- 2 Small areas rather than large are to be preferred as the basis of calculation for Bailey's method.
- 3 Catchment population estimates should be applied cautiously and interpreted carefully.

Appendix 1

An example will be used to illustrate the various methods. The West Midlands Regional Health Authority has been chosen because data was available to the authors, and ophthalmology as the specialty because this was used by Gandy (1979) in a comparison of his own method with Bailey's. A catchment population is calculated for the Wolverhampton Area Health Authority (a single district area) by way of illustration since it has a large catchment population for this specialty.

The relevant data for this example are given in Table 1. It is assumed for the purposes of illustration that no cases from outside the region are dealt with inside the region and that no cases from inside the region are dealt with outside the region. (Hospital activity analysis is organized on a regional basis and the computer files of other regions would, for a full calculation, have to be examined.)

Table 1

Catchment information for the West Midlands Regional Health Authority

Originating Health district	Cases dealt with ^a		Total ($m_1 + m_2$)	Population ^b (n)	Admission ^c rate ($m_1 + m_2$)/ n
	In Wolverhampton (m_1)	Elsewhere (m_2)			
Hereford	1	588	589	145.7	4.043
Broms/Redditch	3	401	404	142.3	2.839
Kidderminster	6	502	508	99.5	5.105
Worcester	1	599	600	222.6	2.695
Salop	93	1 012	1 105	365.9	3.020
Mid Staffs	524	83	607	289.5	2.097
North Staffs	2	888	890	469.1	1.897
S. E. Staffs	89	280	369	238.4	1.548
N. Warks.	8	312	320	169.4	1.889
C. Birmingham	5	603	608	251.5	2.417
E. Birmingham	1	343	344	153.3	2.244
N. Birmingham	47	228	275	168.9	1.628
S. Birmingham	4	630	634	257.2	2.465
W. Birmingham	5	427	432	210.1	2.056
Dudley	278	330	608	296.5	2.050
Sandwell	167	431	598	309.6	1.932
Walsall	545	159	704	265.6	2.651
Wolverhampton	795	8	803	260.6	3.081
WMRHA	2 574	7 824	10 398	4 248.4	2.447

^a Discharges and deaths from Hospital Activity Analysis for 1978.

^b Population in thousands for 1979.

^c Cases per 1 000 population.

Source: West Midlands Regional Health Authority.

Bailey's Method

Ideally we should use smaller geographical units as the basis of calculation; however, using the health district information we have

$$c = \sum_i \{(nm_i)/(m_1 + m_2)\} = 1\,065 \text{ thousands}$$

with (using Bailey's formula for the variance) a standard error of 12.7 thousand.

RAWP/Gandy Method

Following the example of Gandy (1979) we assume in this instance that the expected admission rate is the national admission rate making no allowance for the age-sex structure of the population or morbidity. Gandy's method

under these circumstances is equivalent to what has here been identified as the catchment element in the RAWP calculations (Senn and Shaw, 1978).

Because the region is regarded as being “watertight” we have here

$$z_1 = 8$$

$$z_2 = \sum m_1$$

where the summation is over all health districts except the Wolverhampton HD so that

$$z_2 = 1\,779$$

and

$$r = 260.6$$

(The admission rate for ophthalmology for England and Wales for 1978 is 2.56 per 1 000.)

We thus have

$$c = 260.6 + \frac{1\,779 - 8}{2.56} = 952 \text{ thousand}$$

Under the assumption that z_1 and z_2 have independent Poisson distributions and that the national admission rate d^* can be regarded as fixed then we can estimate variance of the estimate as

$$\sigma_c^2 = \frac{z_1 + z_2}{d^{*2}}$$

so that the standard error is 16.5 thousand.

Cottrell's Method

We have

$$z_1 = 8$$

$$z_2 = 1\,779$$

$$z_3 = 795$$

$$r = 260.6$$

Therefore

$$c = \frac{1\,779 + 795}{795 + 8} \times 260.6 = 835 \text{ thousand}$$

On the assumption that z_1 , z_2 , z_3 are independent Poisson variates an

approximate formula for the variance, using a Taylor's expansion is

$$\sigma_c^2 = \frac{r^2(z_2 + z_3)(z_2 + z_1)}{(z_1 + z_3)^3}$$

so that in this case the standard error is 24·6 thousand

The Median Line Method

Wolverhampton, Mid Staffs, and Walsall all send 50 per cent or more of their cases to Wolverhampton. Although, however, only 46 per cent of all cases from Dudley go to Wolverhampton, this is in fact more than go to any other health district. We thus have a catchment population of

$$c = 260\cdot6 + 289\cdot5 + 265\cdot6 + 296\cdot5 = 1112 \text{ thousand}$$

These results are summarized in Table 2.

Table 2
Comparison of estimated catchment populations

<i>Method</i>	<i>Estimate</i>	<i>Standard error</i>
Bailey	1 065	12·7
RAWP/Gandy	952	16·5
Cottrell	835	24·6
Median line	1 112	—

(All figures in thousands.)

Appendix 2

Bailey's formula for the variance of a catchment population estimate is approximate and as long as it is valid then choosing small areas rather than large areas for catchment estimation does not, as has already been shown, affect the variance. However, the ratio, $m_1/(m_1 + m_2)$ is indeterminate when m_1 and m_2 are equal to zero and this suggests that small levels of admissions may cause some difficulties for small area estimation. This problem was investigated using a computer simulation.

A series of simulations was carried out for a "large" area of 220 000 persons (approximately the average size for a health district in England) (OPCS, 1979) and then for 220 "small" areas each of population 1 000 (the size of a typical parish). It was assumed that expected admission rates in each of the 220 small areas were the same and equal to 1/220th of the expected figure for the large area. Under these circumstances the

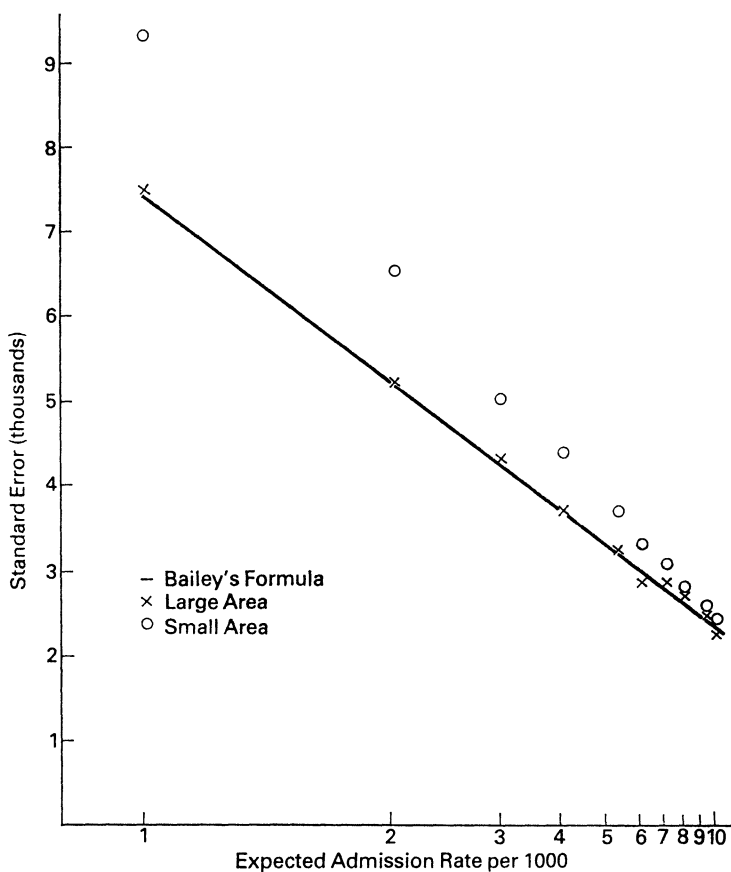


Figure 1

Comparison of Large and Small Area Estimation using Bailey's method when the expected proportion admitted to the "Home" Hospital is 0.5: results of 1 000 simulations

small and large area approaches have nothing to choose between them in terms of bias. The simulation schedule is set out in Table 3.

Thus 30 separate comparisons of small and large area estimation involving 1 000 simulations were run and a further 75 comparisons involving 100 simulations. Admissions were taken to be Poisson, a normal approximation being used where appropriate.

For low expected admission rates for small area simulations it is not unusual to get no admissions at all from a geographical unit. Where this

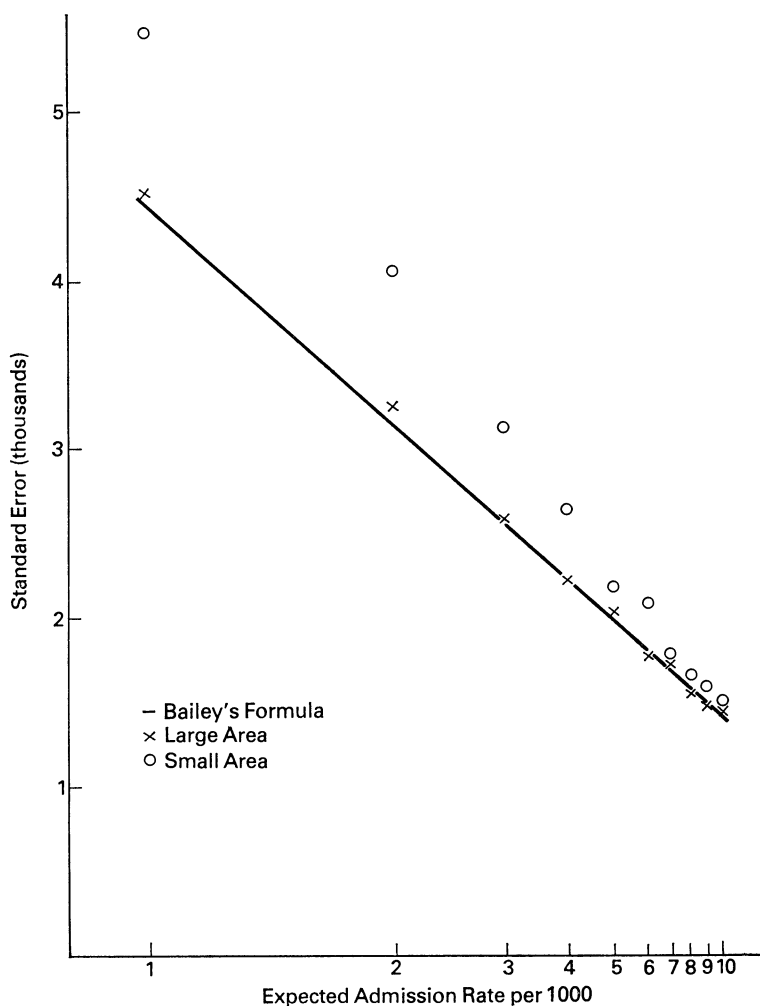


Figure 2
Comparison of Large and Small Area Estimation using Bailey's method
when the expected proportion admitted to the "Home" Hospital is 0.1:
results of 1 000 simulations

occurs the proportion of admissions from the "next" geographical unit is used in the calculation. Thus two or more units will be combined when this is required for estimation purposes.

Figures 1 and 2 show selected results for the simulations. Other things being equal the standard error is inversely proportional to the square

Table 3
Number of catchment simulations

	<i>Expected admission rate per 1 000 (increments of 1)</i>	
	<i>1-10</i>	<i>11-25</i>
Proportion to "home" hospital		
0·1	1 000	100
0·2	—	100
0·3	1 000	100
0·4	—	100
0·5	1 000	100

root of the admission rate and the horizontal scale has been adjusted to reflect this.

Providing the expected level of admissions is at least 10 per unit there is little to choose between the large and small area approach on grounds of variance. However, even with an admission rate as low as 1 per 1 000 the cost in terms of increased variance is small in relation to the sort of bias that can be avoided using small area estimation if there is in fact variation of admission rates across the area.

It is therefore concluded that there is no serious disadvantage in terms of standard error to using small area rather than large area estimation.

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